

Fast Track Jr.

Write a Python program that flies the **CoDrone EDU** through this course **by itself**. You have **20 minutes**. Write and test it in the **Mission Lab** in your browser, submit it, and we'll replay everyone's flight on the big screen.



MISSION LAB

theactualjacob.github.io/fast-track-jr

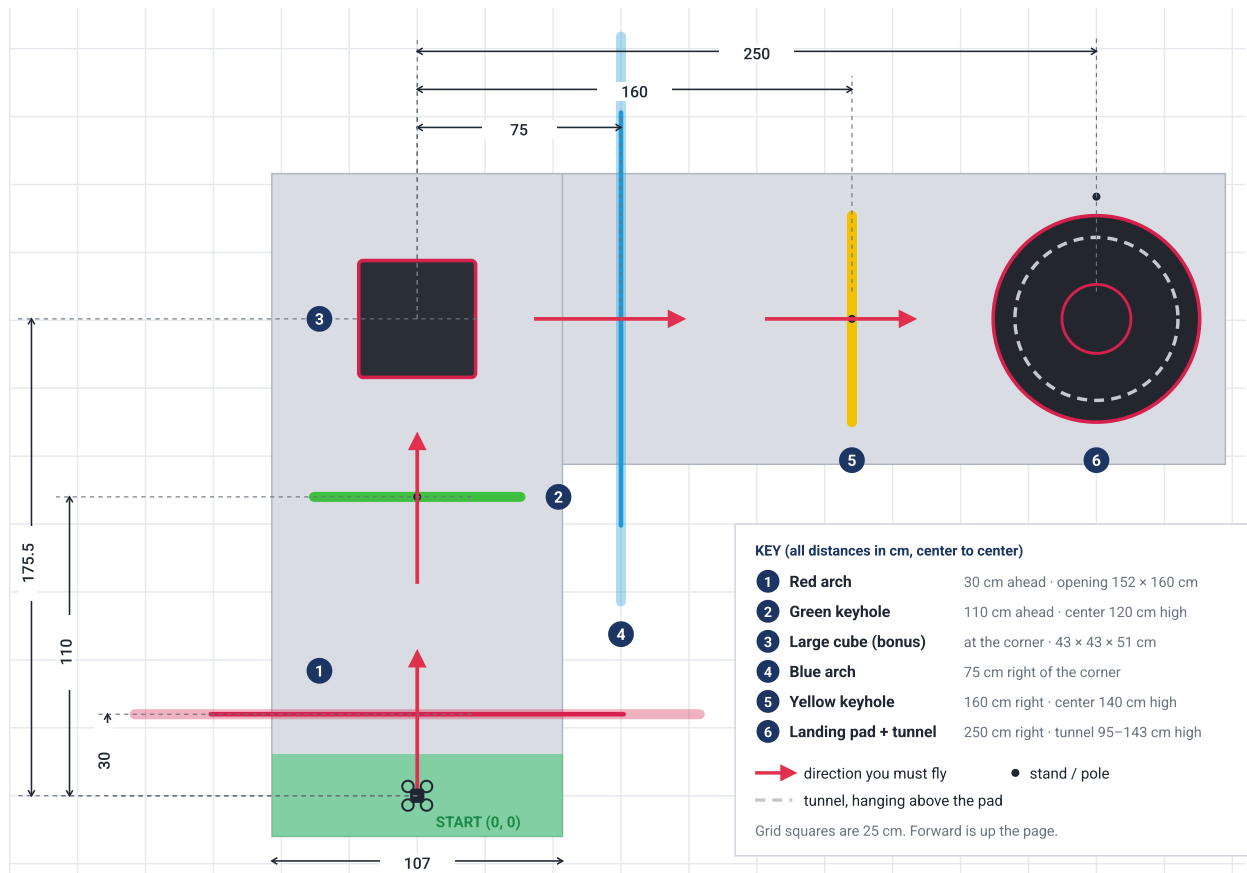

Scoring

Task	Points
Take off	5
Fly under the red arch	5
Fly through the green keyhole	10
Fly through the large cube BONUS	25
Fly under the blue arch	5
Fly through the yellow keyhole	20
Drop through the tunnel BONUS	15
Land on the landing pad	5
Land on top of the cube	10
Land on the bullseye	15

Rules

- The drone starts on the green square, facing the red arch.
- Fly through every gate **in the direction of its arrow**.
- After takeoff() the drone hovers at about **80 cm**.
- Hit anything and you crash.** The run ends, but you keep your points.
- Each gate scores up to 2 times. Only your **final** landing counts, and only one landing option.
- The match is **3 minutes**. Ties go to the faster run.

Course Map



Gate	Where (center)	How high	Fly
Red arch	30 cm forward	opening up to 160 cm	forward
Green keyhole	110 cm forward	center 120 cm	forward
Large cube BONUS	175.5 cm forward (the corner)	holes centered 25.5 cm, 30 cm wide	in one side, out another
Blue arch	corner, then 75 cm right	opening up to 160 cm	right
Yellow keyhole	corner, then 160 cm right	center 140 cm	right
Tunnel BONUS	corner, then 250 cm right	95–143 cm (hangs over the pad)	straight down
Landing pad	corner, then 250 cm right	floor · bullseye 25.4 cm wide	land

Keyhole and tunnel openings are about 60 cm wide and the drone is 14 cm, so aim within ± 20 cm of the center.

Code Cheat Sheet

Every program looks like this

```
from codrone_edu.drone import *

drone = Drone()
drone.pair()

drone.takeoff()      # up to ~80 cm

# your moves go here

drone.land()
drone.close()
```

Which way is which

Moves go the way the drone is **facing** (at the start: toward the red arch). Distances are in **cm**.

forward · toward the red arch

right · toward the second mat

up / down · height

Example: the first two gates

```
drone.move_upward(40)    # 80 → 120 cm
drone.move_forward(140)  # red arch + keyhole
```

Commands

Command	What it does
<code>drone.takeoff()</code>	Take off and hover at about 80 cm.
<code>drone.move_forward(100)</code>	Fly forward 100 cm. Also <code>move_backward</code> .
<code>drone.move_right(50)</code>	Fly sideways 50 cm without turning. Also <code>move_left</code> .
<code>drone.move_upward(40)</code>	Climb 40 cm. Also <code>move_downward</code> .
<code>drone.turn_right(90)</code>	Turn in place 90°. Also <code>turn_left</code> .
<code>drone.hover(1)</code>	Hold still for 1 second.
<code>drone.move_forward(100, "cm", 1)</code>	Third value is the speed in m/s (default 0.5, max 2).
<code>print(drone.get_pos_z())</code>	Print the current height in cm.
<code>drone.land()</code>	Land right where the drone is.

Watch out

- You start at 80 cm and the green keyhole is at 120 cm. **Climb first** or you'll crash.
- Spelling counts: `move_foward` is an error.
- Turns can overshoot a few degrees. `move_right()` is more accurate.
- Always finish with `drone.land()`, or you get no landing points.

Fly & Submit



OPEN THE MISSION LAB

theactualjacob.github.io/fast-track-jr

Runs in your browser. Nothing to install; Chromebooks are fine.

1. Type your **GitHub username** in the top bar (no account? type your name).
2. Fill in the TODOs in the editor, using the course map and the cheat sheet.
3. Press **Fly** (or **Ctrl/⌘ + Enter**). Your flight replays in 3D and the running line lights up. A crash or bug points to its line. Fix it and fly again as many times as you like.
4. Click **Submit**:
 - **Open a pull request** if you have a GitHub account. Then click **Commit changes** → **Propose changes** → **Create pull request**.
 - **Copy share link** if you don't. Paste the link wherever your leader asks.

Hints

Green keyhole	Climb 40 cm before you fly forward.
The corner	It's 175.5 cm ahead. Either <code>turn_right(90)</code> then fly forward, or skip turning and use <code>move_right()</code> .
Yellow keyhole	It's 20 cm higher than the green one (140 cm).
Tunnel over the pad	Safe: drop below 85 cm before you reach it, then land. Bonus: climb above 150 cm, stop right over the pad and <code>land()</code> . You drop through the tunnel for +15, and the bullseye if you're centered.
Large cube (hard)	Holes are 30 cm wide and 25 cm off the floor. Get low before the cube, fly into the middle, then slide out a different side.

After the session every flight is replayed in 3D, all racing at once, with your code on screen. The same file flies the **real** CoDrone EDU, so we'll try the best ones on the real field next.